

Context-Tree Weighting for Real-Valued Time Series

Bayesian Inference with Hierarchical Mixture Models

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Project Overview

Objectives

- Implement the BCT-AR algorithm from the paper by Papageorgiou & Kontoyiannis (2023)
- Extend the framework to a **bivariate** setting
- Evaluate performance against classical AR baselines on both synthetic and real datasets

Models presented

- BCT-AR (Bayesian Context Tree AutoRegressive) — Univariate
- BCT-AR — Bivariate (two variants: V1 and V2)

BCT-AR: Prediction Objective and Quantisation

- **Prediction objective:** Consider a real-valued time series $(x_t)_{t \geq 1}$. We model

$$p(x_t \mid x_{t-1}, x_{t-2}, \dots)$$

for probabilistic forecasting.

- **Quantisation:** Introduce a quantiser $Q : \mathbb{R} \rightarrow \mathcal{A} = \{0, \dots, m-1\}$:

$$Q(x) = \begin{cases} 0, & x < c_1, \\ i, & c_i \leq x \leq c_{i+1}, \\ m-1, & x > c_{m-1}. \end{cases}$$

Maps real-valued observations to a finite alphabet.

- **Discrete context:** For maximum depth D , define the quantised past

$$t_t = (Q(x_{t-1}), \dots, Q(x_{t-D}))$$

encoding the recent history.

BCT-AR: Context Tree and State Partition

- **Context tree:** A proper m -ary tree T of depth at most D . The context s_t of x_t is the leaf of T that is a suffix of t_t , allowing variable-length memory.
- **State partition:** The leaves $s \in T$ define a partition of past histories into *regimes*, each with a different predictive behaviour.
- **General model (BCT-SSM):** A parametric model M_s with parameters θ_s is assigned to each leaf. The likelihood factorises as

$$p(x \mid T, \theta) = \prod_{s \in T} \prod_{i \in B_s} p(x_i \mid \theta_s, x_{i-1}, \dots, x_{i-D}),$$

where B_s is the set of indices assigned to context s .

BCT-AR: Bayesian Prior Structure

- **Tree prior:** A Bayesian prior is placed over trees:

$$\pi(T) = \alpha^{|T|-1} \beta^{|T|-L_D(T)}, \quad \alpha = (1 - \beta)^{1/(m-1)},$$

penalising complex trees and controlling overfitting.

- **Parameter prior:** Conditional on T , parameters are independent across leaves:

$$\pi(\theta \mid T) = \prod_{s \in T} \pi(\theta_s).$$

- **Evidence (marginal likelihood):**

$$p(x) = \sum_T \pi(T) \int p(x \mid T, \theta) \pi(\theta \mid T) d\theta$$

integrates over all trees and parameters.

- **Node marginal likelihood:** For each node s :

$$P_e(s, x) = \int \prod_{i \in B_s} p(x_i \mid \theta_s) \pi(\theta_s) d\theta_s.$$

BCT-AR: CCTW and CBCT Algorithms

CCTW recursion (evidence computation)

$$P_{w,s} = \begin{cases} P_e(s, x) & \text{(leaf),} \\ \beta P_e(s, x) + (1 - \beta) \prod_{j=0}^{m-1} P_{w,sj} & \text{(otherwise).} \end{cases}$$

At the root, $P_{w,\text{root}} = p(x)$ exactly.

CBCT recursion (MAP tree)

$$P_{m,s} = \max \left\{ \beta P_e(s, x), (1 - \beta) \prod_{j=0}^{m-1} P_{m,sj} \right\}$$

Produces the MAP tree $T^* = \arg \max_T p(T \mid x)$ via a pruning procedure.

BCT-AR: The Autoregressive Leaf Model

- **BCT-AR model:** Each leaf uses an $\text{AR}(p)$ model:

$$x_t = \phi_s^\top \tilde{x}_{t-1} + e_t, \quad e_t \sim \mathcal{N}(0, \sigma_s^2), \quad \tilde{x}_{t-1} = (x_{t-1}, \dots, x_{t-p})^\top.$$

Results in a nonlinear mixture of local linear models.

- **Conjugate priors:**

$$\sigma_s^2 \sim \text{Inv-Gamma}(\tau, \lambda), \quad \phi_s \mid \sigma_s^2 \sim \mathcal{N}(\mu_0, \sigma_s^2 \Sigma_0).$$

- **Sufficient statistics:**

$$s_1 = \sum_{i \in B_s} x_i^2, \quad s_2 = \sum_{i \in B_s} x_i \tilde{x}_{i-1}, \quad s_3 = \sum_{i \in B_s} \tilde{x}_{i-1} \tilde{x}_{i-1}^\top.$$

BCT-AR: Closed-Form Posteriors and MAP Estimates

- **Integrated likelihood (closed form):**

$$P_e(s, x) = C_s^{-1} \frac{\Gamma(\tau + |B_s|/2) \lambda^\tau}{\Gamma(\tau) (\lambda + D_s/2)^{\tau + |B_s|/2}}.$$

- **Posterior distributions:**

$$\sigma_s^2 \mid T, x \sim \text{Inv-Gamma}\left(\tau + \frac{|B_s|}{2}, \lambda + \frac{D_s}{2}\right),$$

$$\phi_s \mid T, x \sim t_\nu(m_s, P_s).$$

- **MAP estimates (used for prediction):**

$$\hat{\phi}_s = m_s, \quad \hat{\sigma}_s^2 = \frac{2\lambda + D_s}{2\tau + |B_s| + 2}.$$

- **Sequential updates:** When a new observation arrives, only $D + 1$ nodes need to be updated $\Rightarrow O(D)$ operations per step (i.e. $O(1)$ in n).

Univariate BCT-AR: Implementation

One BCT-AR class with two main methods:

observe(x)

- 1 Quantise the past of x up to max depth $\Rightarrow$ path in the tree.
- 2 Traverse the tree along the path; update $\{|B_s|, s_1, s_2, S_3\}$ and $P_e(s, x)$. Update AR MAP parameters. (*Complexity: linear in depth.*)
- 3 Apply CCTW update: recursive update of $P_{w,s}$.
- 4 Apply CBCT update: recursive update and pruning of $P_{m,s}$.

predict(x)

- 1 Quantise the context up to max depth $\Rightarrow$ path in the tree.
- 2 Traverse the tree until a leaf is reached.
- 3 Predict the next value using the leaf's MAP parameters.

Key advantages: Captures nonlinear dependencies; computationally efficient.

Univariate BCT-AR: Computational Efficiency

Comparison with simple AR model from the statsmodels Python package.

Time complexity and error in AR coefficients estimation using statsmodels package

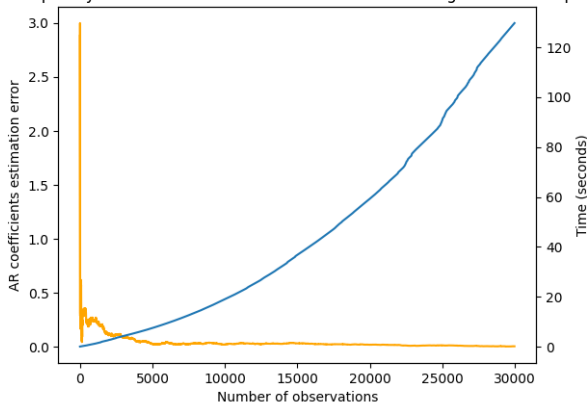


Figure: Complexity: statsmodels AR model (sequential refitting)

Univariate BCT-AR: Computational Efficiency (cont.)

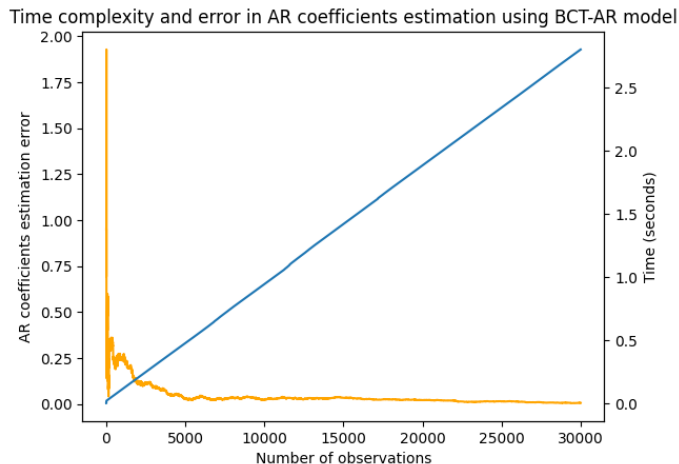


Figure: Complexity: BCT-AR model (sequential update)

The BCT-AR model supports efficient online updates, unlike standard AR refitting.

Univariate: Artificial Data — Setup

- First tested on artificially generated data with increasing complexity.
- Simple AR(2) test:

$$x_t = a x_{t-1} + b x_{t-2} + \sigma$$

The algorithm trivially recovered the model, so we moved to harder settings.

We worked with AR(2) mixtures conditioned on the past of the variable:

$$x_t = \begin{cases} a_1 x_{t-1} + b_1 x_{t-2} + \sigma_1 & \text{if } x_{t-1} < 0, \\ a_2 x_{t-1} + b_2 x_{t-2} + \sigma_2 & \text{if } x_{t-1} \geq 0. \end{cases}$$

Univariate: Artificial Data — Results

Training on 20 000 data with

$$a_1 = 0.5, b_1 = -0.3, \sigma_1 = 0.1, a_2 = -0.4, b_2 = 0.2, \sigma_2 = 0.2.$$

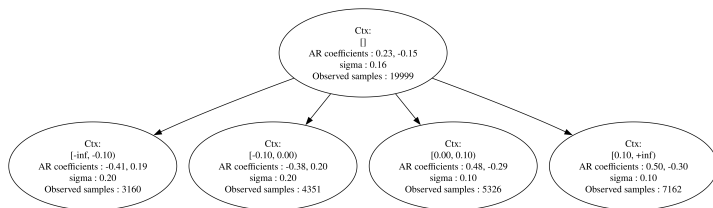


Figure: Univariate tree for AR mixture

The algorithm correctly identifies the two regimes and prunes the tree to depth 1 (max depth was 3), understanding that older context is not relevant.

Univariate: Sensitivity to Hyperparameters

However, the model is sensitive to the number of training samples and β . Setting the number of training samples to only 100 leads to this underfitting (averaged) tree:

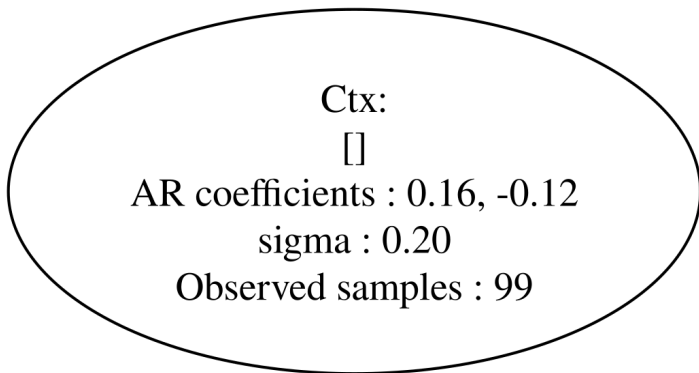


Figure: Underfitting with univariate tree for AR mixture (100 samples)

Conclusion: Architecture performs well on shallow artificial data.

Univariate: Real Data — Unemployment Rate Dataset

- US unemployment rate from 1948 to 2018.
- Following the paper: work on the first-differenced series $\Delta x_t = x_t - x_{t-1}$.
- Threshold set at 0.15 (splits nodes into > 0.15 and ≤ 0.15).
- Dataset standardised.

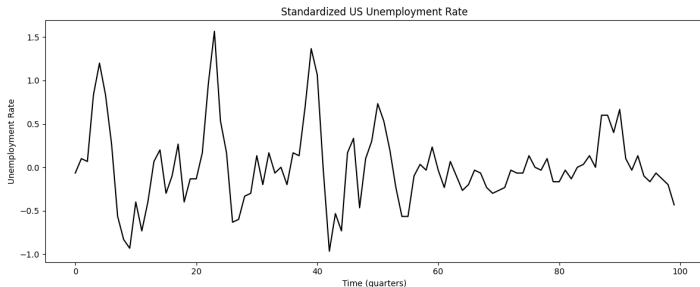


Figure: US unemployment rate (no regimes)

Univariate: Real Data — Fitted Tree

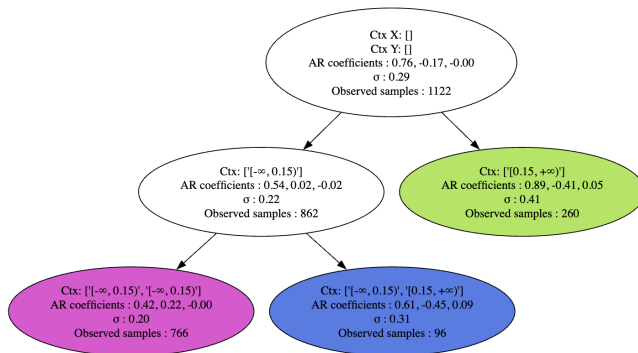


Figure: BCT-AR fitted tree on US unemployment rate

Univariate: Real Data — Regime Identification

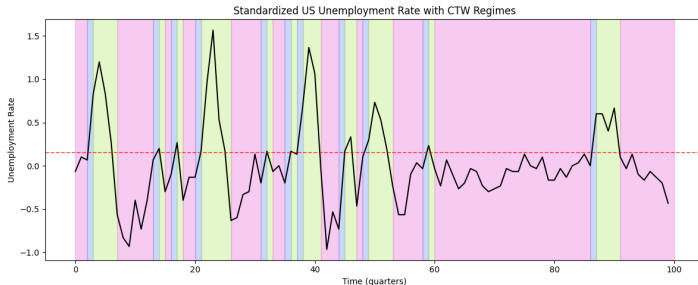


Figure: Unemployment rate with identified regimes over time

Univariate: Real Data — Forecasting Results

1-step prediction results on the US unemployment rate dataset:

Model	MSE	MAE
Baseline (lag 1)	0.10323	0.23618
AR(2)	0.08268	0.21389
CTW (BCT-AR)	0.07758	0.20355

Table: 1-step forecasting results on US unemployment rate

BCT-AR achieves the best performance on both MSE and MAE.

Bivariate BCT-AR: Motivation and Ideas

Idea: Add additional data to the context to improve prediction (from intuition or correlated data).

Bivariate V1

Change the quantiser to quantise along two axes simultaneously at each level.

Issues: Very small slots per node (few observations), prone to overfitting.

Bivariate V2

Configured with a *quantiser sequence* indicating the order of quantisation variables (e.g. $[1, 0, 0]$). Unidimensional quantisers are passed for each variable.

- **Goal:** Allow splitting on X more than on Y — each depth level corresponds to a split on a given variable.
- The tree depth penalisation naturally discourages over-splitting on Y .

Bivariate: Artificial Data — Setup

We generated a series x from a mixture of AR models, conditioned on the value of another series y (itself an AR process):

$$x_t = \begin{cases} a_1 x_{t-1} + b_1 x_{t-2} + \sigma_1 & \text{if } y_t < \theta_1, \\ a_2 x_{t-1} + b_2 x_{t-2} + \sigma_2 & \text{if } \theta_1 \leq y_t < \theta_2, \\ a_3 x_{t-1} + b_3 x_{t-2} + \sigma_3 & \text{if } \theta_2 \leq y_t < \theta_3, \\ a_4 x_{t-1} + b_4 x_{t-2} + \sigma_4 & \text{if } y_t \geq \theta_3. \end{cases} \quad (1)$$

$$y_t = ay_{t-1} + by_{t-2} + \sigma.$$

Both bivariate models were trained on the same data with the same hyperparameter tuning procedure.

Bivariate: Artificial Data — Parameters and V1 Results

Parameters:

$$\sigma_{1,2,3,4} = 0.05, \quad a_1=1, \quad b_1=-0.5, \quad a_2=-0.8, \quad b_2=0.7, \quad a_3=0.4, \quad b_3=-0.7, \\ a_4=-0.5, \quad b_4=0.4, \quad \theta_1=-0.05, \quad \theta_2=0, \quad \theta_3=0.05.$$

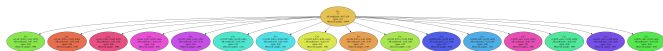


Figure: Bivariate tree (V1) for conditioned AR mixture

The model correctly identifies the coefficients and prunes unnecessary depth (max depth = 3).

Bivariate: Artificial Data — V2 Results



Figure: Bivariate tree (V2) for conditioned AR mixture

Different tree shape but consistent results: the model correctly identifies the underlying structure.

Bivariate: Artificial Data — Prediction Benchmark

Prediction at horizon $k \geq 1$. High signal-to-noise ratio of AR models ($\approx 50\%$) makes long-horizon prediction difficult.

Model	MAE ($k = 1$)	MAE ($k = 10$)
Baseline (lag 1)	0.129 ± 10^{-2}	$0.155 \pm 2 \times 10^{-2}$
AR	$0.072 \pm 9 \times 10^{-3}$	$0.090 \pm 8 \times 10^{-3}$
Bivariate tree V1	$0.068 \pm 9 \times 10^{-3}$	$0.085 \pm 9 \times 10^{-3}$
Bivariate tree V2	$0.064 \pm 8 \times 10^{-3}$	$0.092 \pm 8 \times 10^{-3}$

Table: Benchmark on MAE for two prediction horizons k

Bivariate trees outperform shallow baselines; improvement over AR is modest at longer horizons due to noise.

Conclusion on artificial data: Both architectures perform well on generated data, but the task is either too easy or too random for strong differentiation.

⇒ We apply the models to **real datasets** to solve difficult, realistic tasks and highlight the bivariate tree's relevance in practice.

Real Data: Météo Dataset — Overview

- Temperature in Île-de-France and electricity consumption.
- Working on the 24h-differenced data.

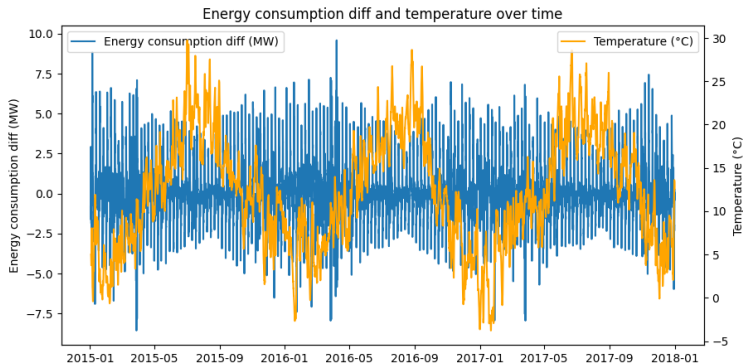


Figure: Overview of temperature and electricity consumption data

Real Data: Météo Dataset — Regime Detection

Grid search on quantiser thresholds found the best parameters. The model was allowed to split on X as well, but stopped before doing so, confirming that no split on X is meaningful.

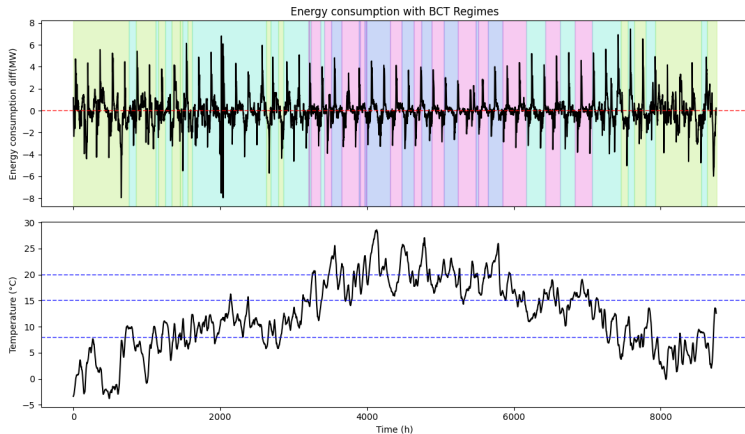


Figure: Observed regimes over a full year

Real Data: Météo Dataset — Fitted Tree

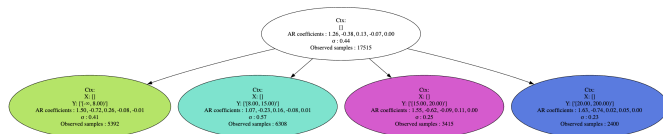


Figure: BCT-AR bivariate tree with identified regimes

For high and low temperatures, similar models emerge (air-conditioning and heating produce the same behaviour).

Real Data: Météo Dataset — Results

Model	MSE	MAE
Baseline (lag 1)	0.17269	0.22894
AR(4)	0.17923	0.21099
CTW (BCT-AR bivariate)	0.13664	0.20729

Table: 1-step forecasting results on Météo dataset

BCT-AR bivariate achieves the lowest MSE and MAE, outperforming both the naive baseline and AR(4).

Real Data: Arrhythmia Dataset — Overview

- Two ECG signals: **MLII** (classic ECG lead: potential difference between right arm and left leg; very sharp peaks) and **V5** (lateral view of the heart; slightly different waveform).
- Both measure cardiac electrical activity from two different angles.

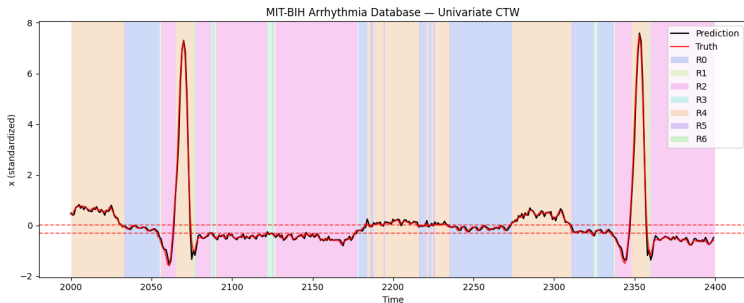


Figure: Arrhythmia dataset: true vs predicted series (univariate model)

Real Data: Arrhythmia Dataset — Bivariate Results

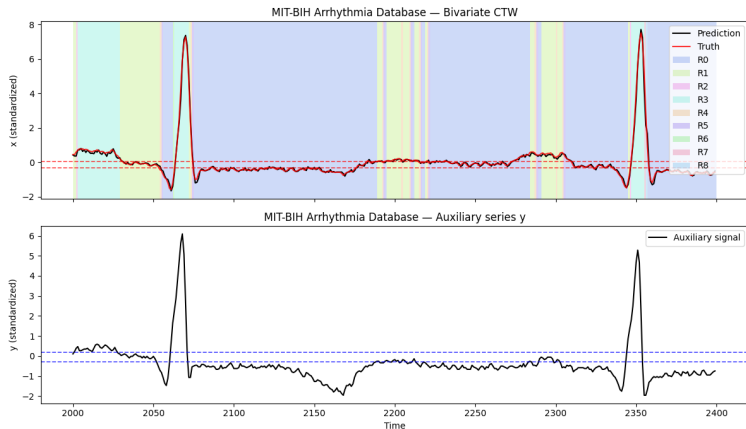


Figure: True vs predicted series (bivariate model)

The bivariate model shows very similar regime zones to the univariate, with a few exceptions where instabilities are resolved.

Real Data: Arrhythmia Dataset — Quantitative Results

Model	MSE ($k = 1$)	MAE ($k = 1$)	MSE ($k = 5$)	MAE ($k = 5$)
Baseline lag1	0.100979	0.119317	0.856994	0.249120
AR(2)	0.019845	0.091185	0.170700	0.255355
Univariate BCT-AR	0.016130	0.080814	0.094927	0.167314
Bivariate BCT-AR	0.011590	0.074459	0.296630	0.200757

Table: Forecasting results on Arrhythmia dataset

The bivariate model achieves the best 1-step MSE and MAE. At horizon $k = 5$, the univariate model is stronger on MSE, while bivariate leads on MAE.

Real Data: PPG Dataset — Overview and Univariate Results

- Patients in intensive care. **X**: PPG signal (variation in blood volume in micro-vessels driven by each heartbeat). **Y**: patient respiration.
- Respiration modifies thoracic pressure, cardiac output and therefore PPG.

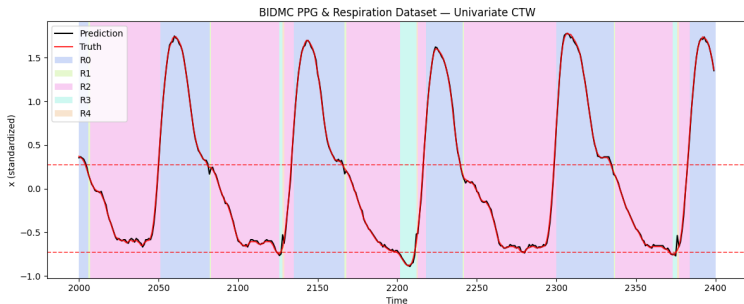


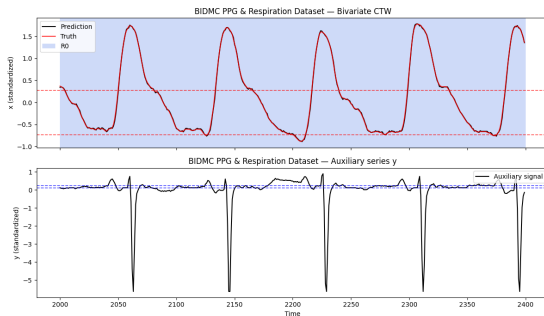
Figure: PPG dataset: univariate model with detected regime changes

The model correctly detects regime changes and deems them relevant.

Real Data: PPG Dataset — Results

Model	MSE ($k = 1$)	MAE ($k = 1$)	MSE ($k = 5$)	MAE ($k = 5$)
Baseline lag1	0.008626	0.062197	0.088087	0.198508
AR	0.000396	0.015423	0.012666	0.071024
Univariate BCT-AR	0.000549	0.015877	0.017541	0.081666
Bivariate BCT-AR	0.000409	0.015379	0.010900	0.071941

Table: Forecasting results on PPG dataset

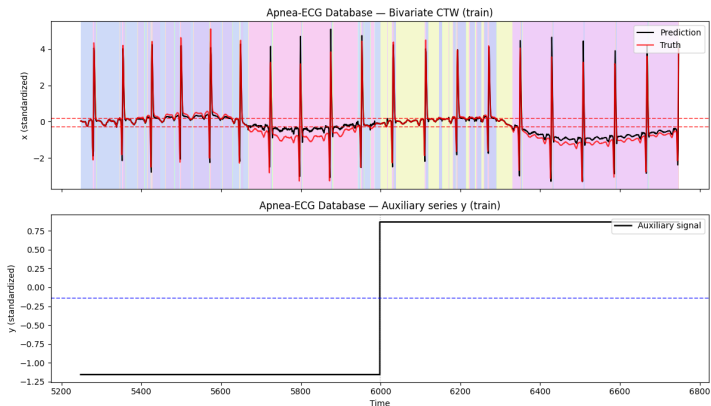


Real Data: PPG Dataset — Discussion

- AR is competitive here; the bivariate model gave just a simple AR tree, demonstrating that the model can stop at an AR when appropriate.
- Bivariate BCT-AR edges out plain AR on MSE at $k = 5$ and on MAE at $k = 1$, showing a marginal but consistent improvement.

Real Data: Sleep Apnea Dataset — Setup

- **X**: ECG time series to predict. **Y**: binary label $y \in \{0, 1\}$ indicating whether the patient is experiencing sleep apnea.
- Goal: condition on a *binary indicator* rather than a continuous series, to force regimes corresponding to apnea vs. no apnea.
- Bivariate V2 used with quantiser sequence $[1, 0, 0, 0]$: forces quantisation on y first (binary case), then the model chooses further splits.



Real Data: Sleep Apnea Dataset — Fitted Tree



Figure: Bivariate BCT-AR tree (first split on y , the apnea indicator)

Real Data: Sleep Apnea Dataset — Quantitative Results

Model	MSE ($k = 1$)	MAE ($k = 1$)	MSE ($k = 5$)	MAE ($k = 5$)
Baseline lag1	0.356286	0.193859	0.718213	0.246023
AR	0.181481	0.152179	0.454537	0.281735
Univariate BCT-AR	0.108016	0.099439	0.376210	0.221578
Bivariate BCT-AR	0.145216	0.123064	0.557362	0.250039

Table: Forecasting results on Sleep Apnea dataset

- Here the bivariate model underperforms the univariate, but still outperforms plain AR.
- The binary conditioning on y does not provide sufficient additional predictive structure.

Conclusion

Main contributions

- Implemented the BCT-AR algorithm from a recent paper for time series prediction/compression.
- Proved the relevance of this model compared to classical autoregressive approaches.
- Showed that the BCT-AR models do *not* work homogeneously: they can significantly outperform AR models in certain setups (e.g. unemployment, arrhythmia, météo).
- Extended the framework to a bivariate setting (two variants: V1, V2).

Observations

- Models are highly sensitive to initialisation and hyperparameters.
- A rigorous method for hyperparameter selection would help stabilise training.

Perspectives and Open Questions

- **Compression:** We did not focus much on compression. Although the link with prediction is well-documented, it would be interesting to extend the current algorithm for time series compression.
- **Deep learning comparison:** We compared with autoregressive models. Given the rise of deep learning, comparing BCT-AR with attention-based neural networks (e.g. Transformers) would be a natural next step.
- **Hyperparameter selection:** Developing a more principled, data-driven approach to select β , quantiser thresholds, and training window size.
- **Bivariate extensions:** Exploring more structured conditioning variables (e.g. exogenous covariates, latent states) could further improve the bivariate model's usefulness.

Thank you for your attention.

Questions?